

IN THE NAME OF ALLAH

THE MOST BENEFICENT AND MERCIFUL

Operation Theatre Assistive Robot (OTAR)

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INTRODUCTION

Assistive Robot

- ❖ Hand detection
- ❖ Voice Recognition
- ❖ Robotic Arm



PROBLEM STATEMENT & OBJECTIVES

Problem statement:

In Operation theatre, surgeons perform many types of complex procedures and due to repetition of tasks; there are chances of occurrence of error. Reasons are:

- ❖ Human error or mishandling of surgical tools
- ❖ Lack of staff

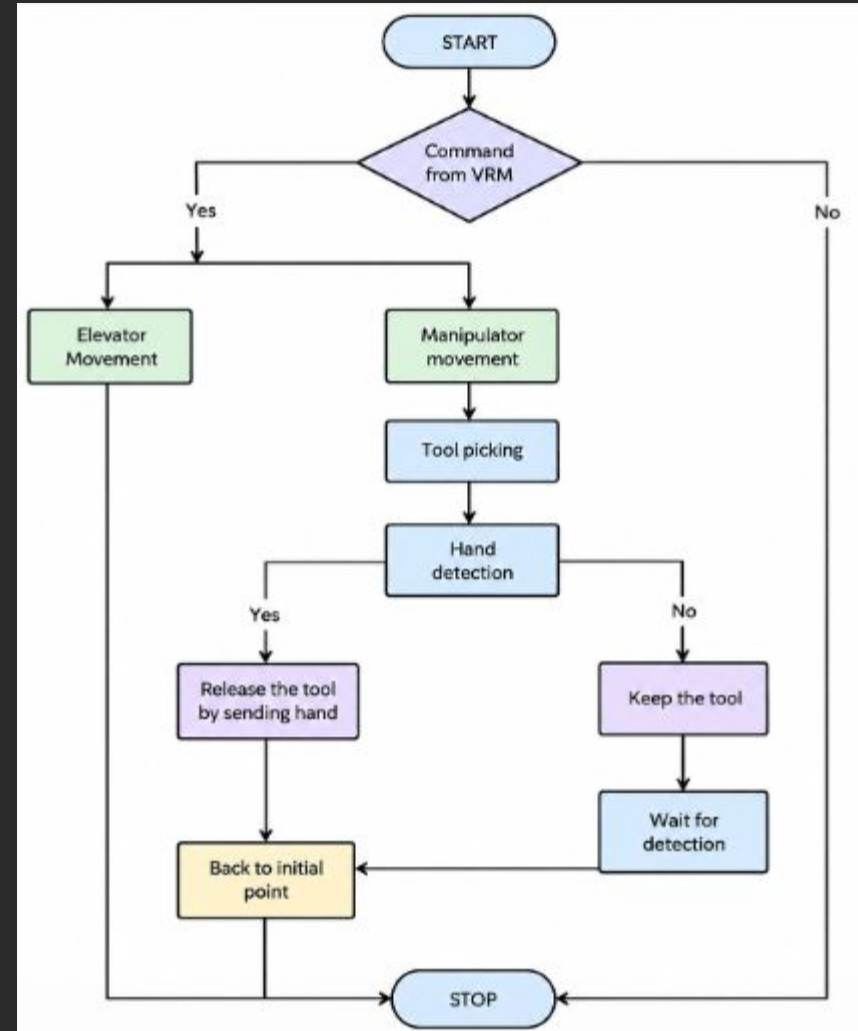
Objectives:

- ❖ Design and Fabrication of robotic arm.
- ❖ Hand detection by Color Recognition.
- ❖ Restricted Voice Recognition.
- ❖ Coupling of all three mentioned above.

METHODOLOGY

Steps:

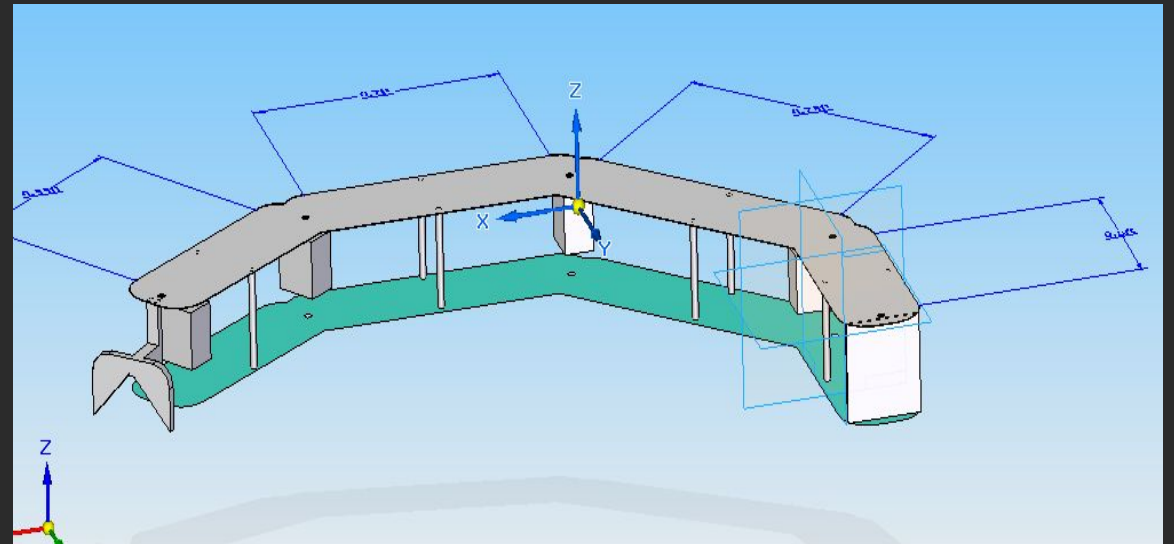
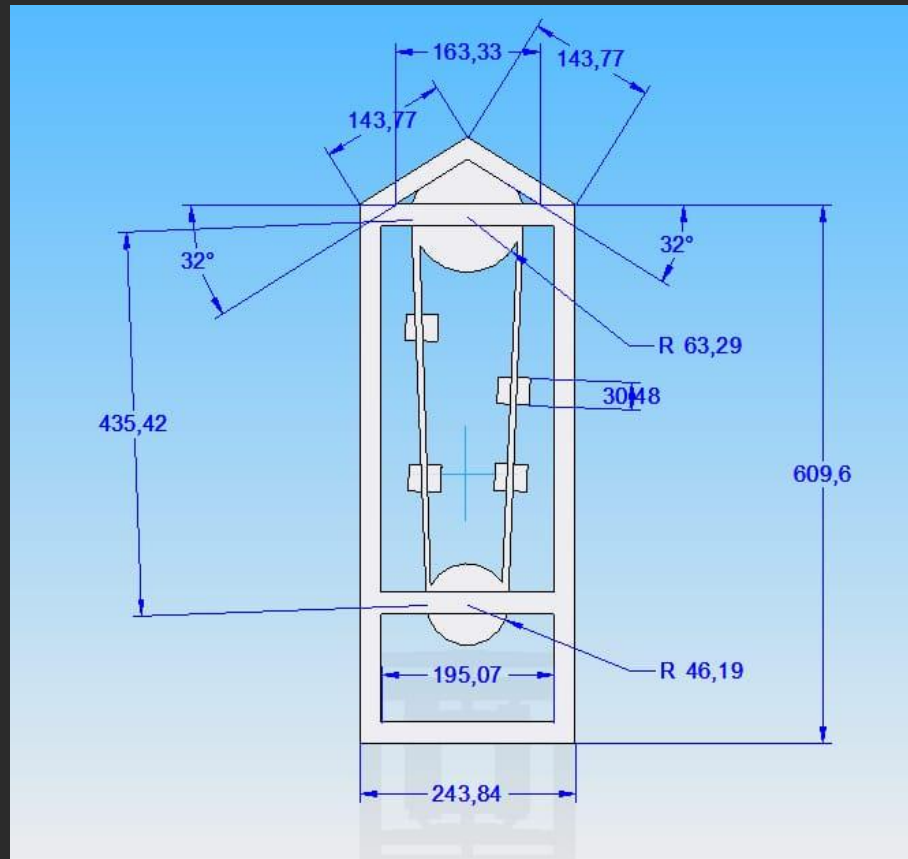
- ❖ Recognizing the voice of surgeon
- ❖ Manipulator and elevator will move simultaneously
- ❖ Picking the tool
- ❖ Hand detection
- ❖ Release the tool
- ❖ Back to initial position



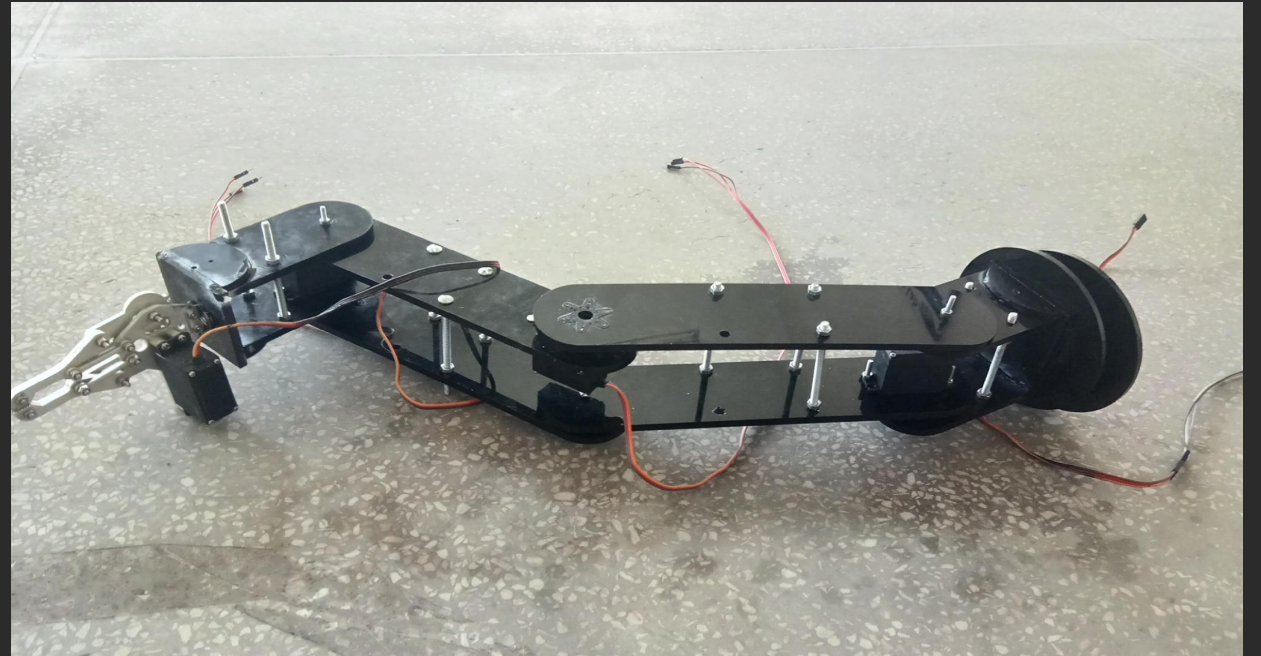
MATHEMATICAL CALCULATIONS

Workspace	(1.5x1.5) ft
Height of operating table	2- 2.5 ft [6]
Height of elevator	3.2 ft
Width of elevator	1.3 ft
Diameter of top gear	19 cm
Diameter of bottom gear	8 cm
Total length of manipulator	2.5 ft
Payload	250 g

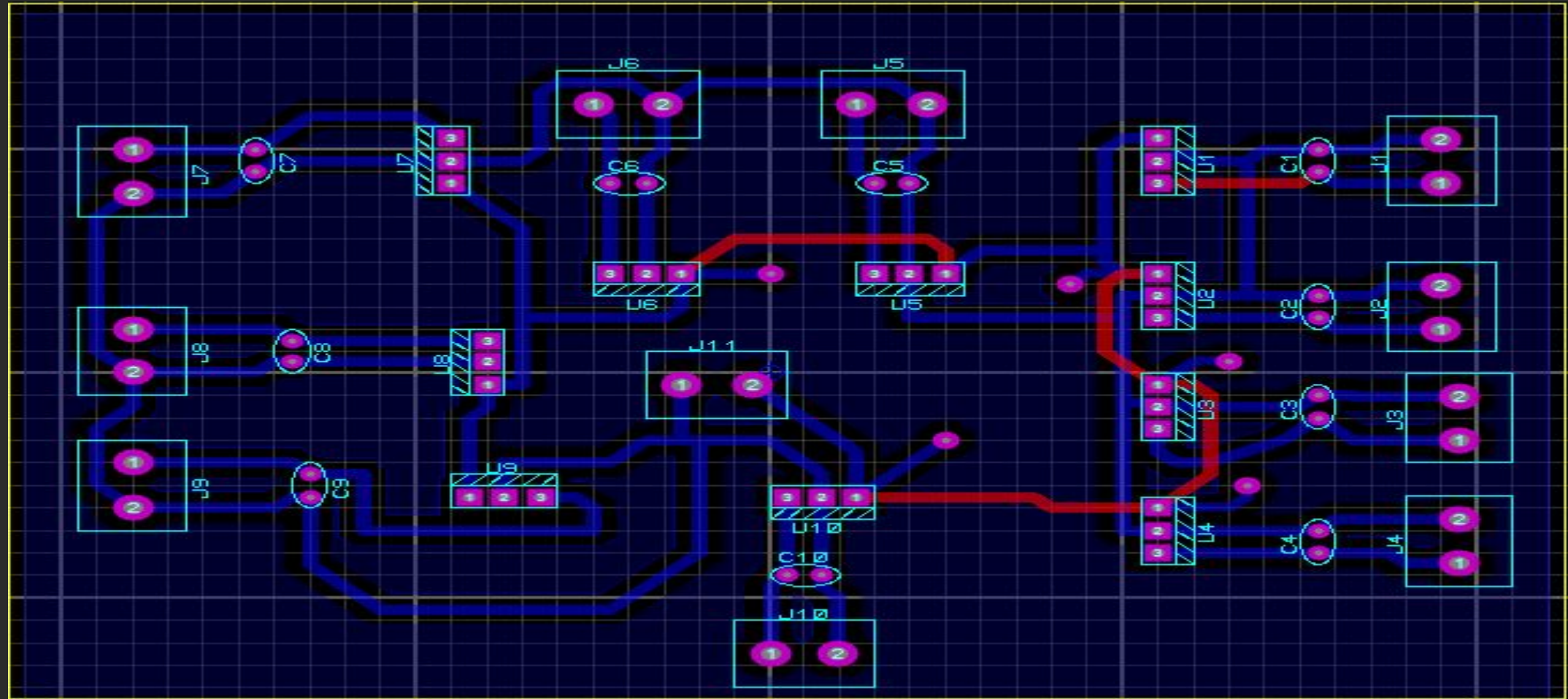
CAD MODEL



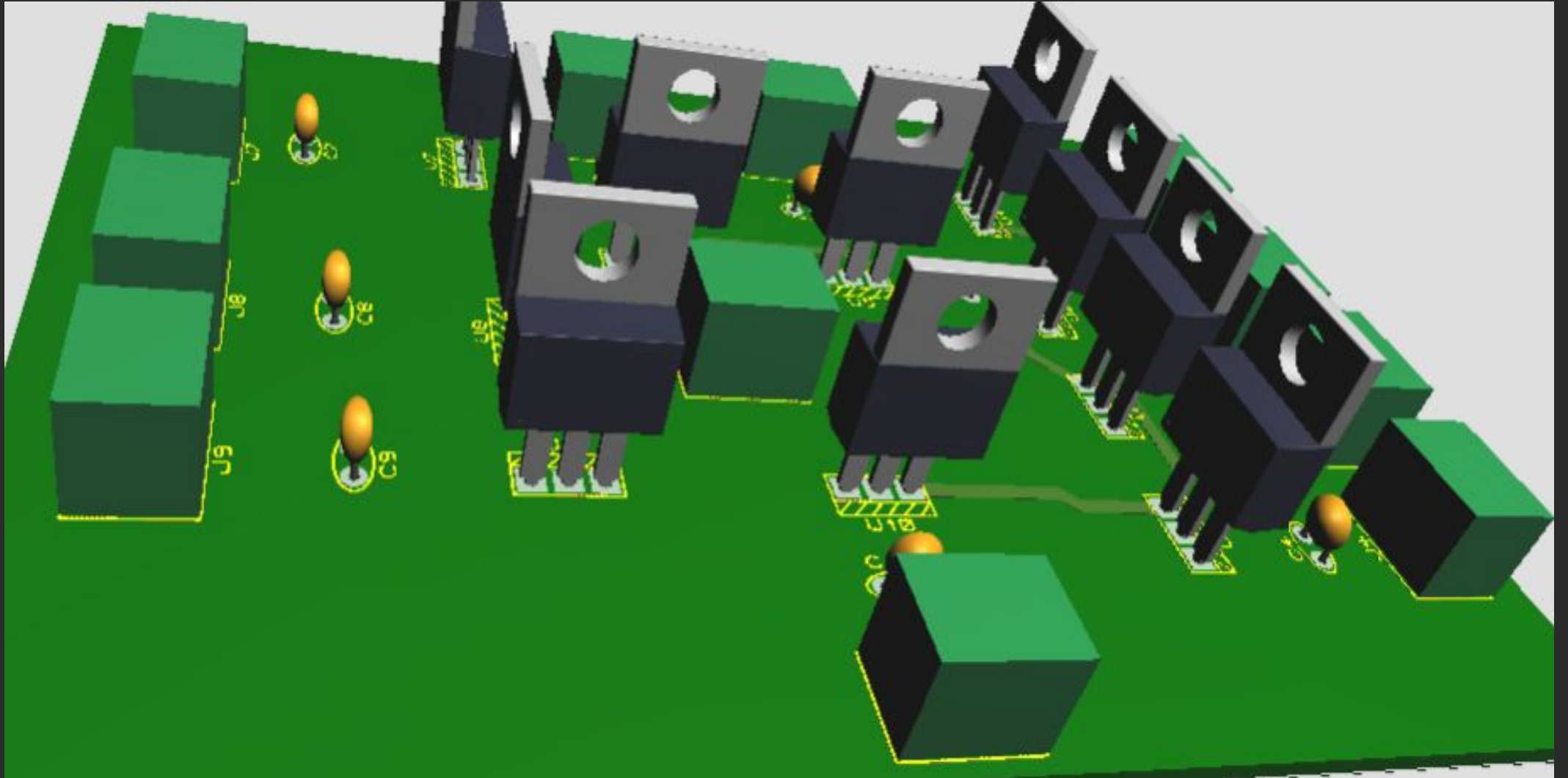
MECHANICAL MODELING



CONVERTER CIRCUIT

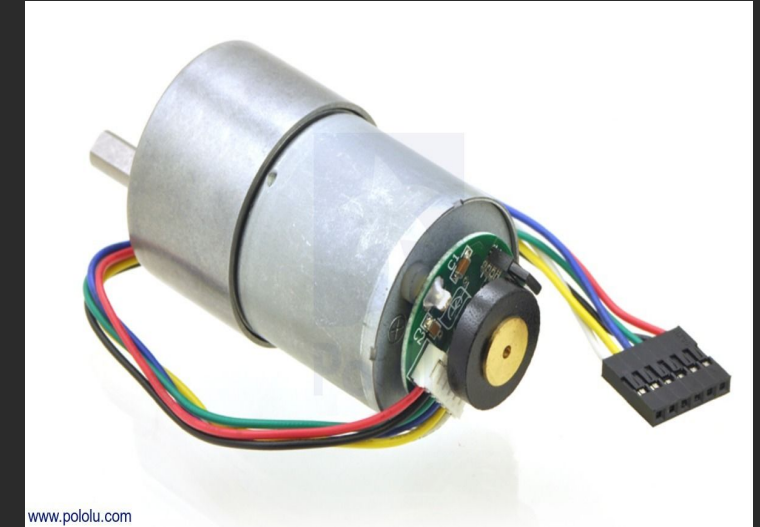
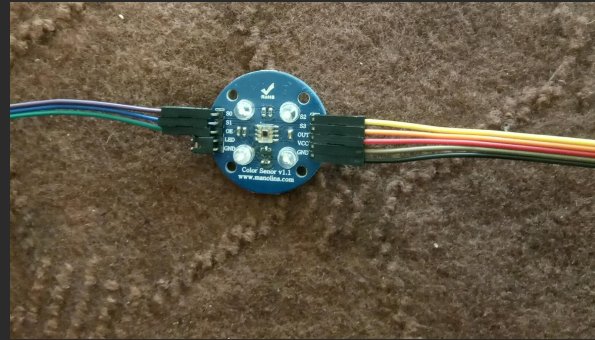


CONVERTER DESIGN



ELECTRICAL COMPONENTS

- ❖ Geetech Module
- ❖ Color Sensor V1.1
- ❖ Servo Motor (MG-996R)
- ❖ Arduino (UNO, Mega)
- ❖ Encoded Motor



VOICE RECOGNITION ALGORITHM

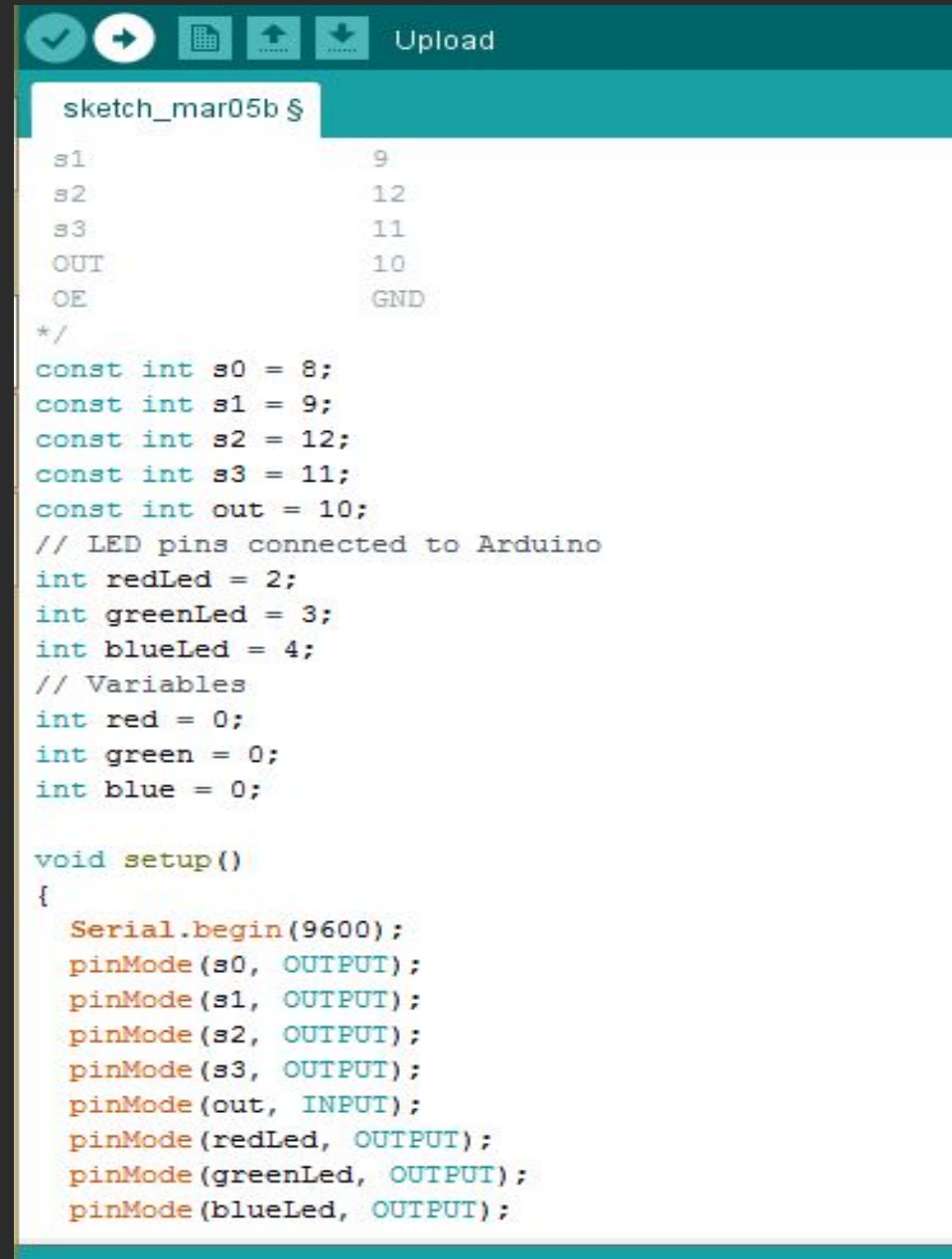
```
vr_sample_train

* train      train (r0) (r1)...      train 0 2 45      Train records
* load       load (r0) (r1) ...      load 0 51 2 3      Load records
* clear      clear                  clear                remove all records in Recognizer
* record     record / record (r0) (r1)... record / record 0 79 Check record train status
* vr         vr                    vr                   Check recognizer status
* getsig     getsig (r)             getsig 0            Get signature of record (r)
* sigtrain   sigtrain (r) (sig)      sigtrain 0 ZERO    Train one record(r) with signature(sig)
* settings   settings              settings             Check current system settings
*****
* @section HISTORY
*
* 2013/06/13   Initial version.
*/
#include <SoftwareSerial.h>
#include "VoiceRecognitionV3.h"
/**
```

```
sketch_jan07b

int ret;
ret = myVR.recognize(buf, 50);
if(ret>0){
  switch(buf[1]){
    case onRecord:
      /** turn on LED */
      digitalWrite(12, HIGH);
      digitalWrite(11, HIGH);
      break;
    case offRecord:
      /** turn off LED*/
      digitalWrite(12, HIGH);
      digitalWrite(11, LOW);
      break;
    case inRecord:
      /** turn on LED */
      digitalWrite(11, HIGH);
      digitalWrite(12, LOW);
      break;
    case outRecord:
      /** turn on LED */
      digitalWrite(12, LOW);
      digitalWrite(11, LOW);
      break;
    default:
      Serial.println("Record function undefined");
      break;
  }
  /** voice recognized */
  printVR(buf);
}
```

HAND DETECTION ALGORITHM



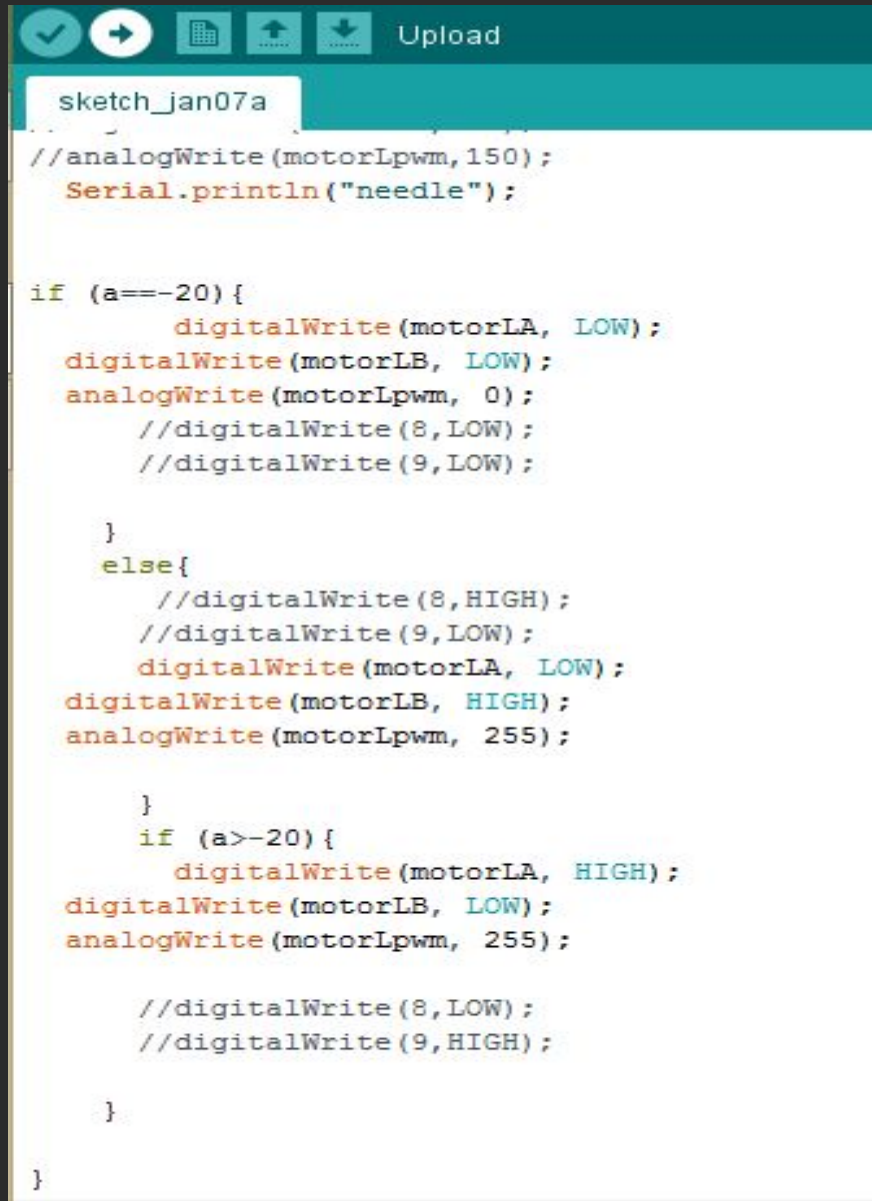
The image shows a screenshot of an Arduino IDE interface. At the top, there is a teal header bar with icons for a checkmark, a right arrow, a grid, an upload arrow, and a download arrow, followed by the text "Upload". Below this is a tab labeled "sketch_mar05b \$". The main area contains the following C++ code:

```

s1      9
s2      12
s3      11
OUT     10
OE      GND
*/
const int s0 = 8;
const int s1 = 9;
const int s2 = 12;
const int s3 = 11;
const int out = 10;
// LED pins connected to Arduino
int redLed = 2;
int greenLed = 3;
int blueLed = 4;
// Variables
int red = 0;
int green = 0;
int blue = 0;

void setup()
{
  Serial.begin(9600);
  pinMode(s0, OUTPUT);
  pinMode(s1, OUTPUT);
  pinMode(s2, OUTPUT);
  pinMode(s3, OUTPUT);
  pinMode(out, INPUT);
  pinMode(redLed, OUTPUT);
  pinMode(greenLed, OUTPUT);
  pinMode(blueLed, OUTPUT);
}
```

ELEVATOR MECHANISM ALGORITHM



```

✓ → 📁 ⬆️ ⬇️ Upload
sketch_jan07a
//analogWrite (motorLpwm, 150);
Serial.println("needle");

if (a== -20) {
    digitalWrite (motorLA, LOW);
    digitalWrite (motorLB, LOW);
    analogWrite (motorLpwm, 0);
    //digitalWrite (8, LOW);
    //digitalWrite (9, LOW);

}
else{
    //digitalWrite (8, HIGH);
    //digitalWrite (9, LOW);
    digitalWrite (motorLA, LOW);
    digitalWrite (motorLB, HIGH);
    analogWrite (motorLpwm, 255);

}
if (a> -20) {
    digitalWrite (motorLA, HIGH);
    digitalWrite (motorLB, LOW);
    analogWrite (motorLpwm, 255);

    //digitalWrite (8, LOW);
    //digitalWrite (9, HIGH);

}

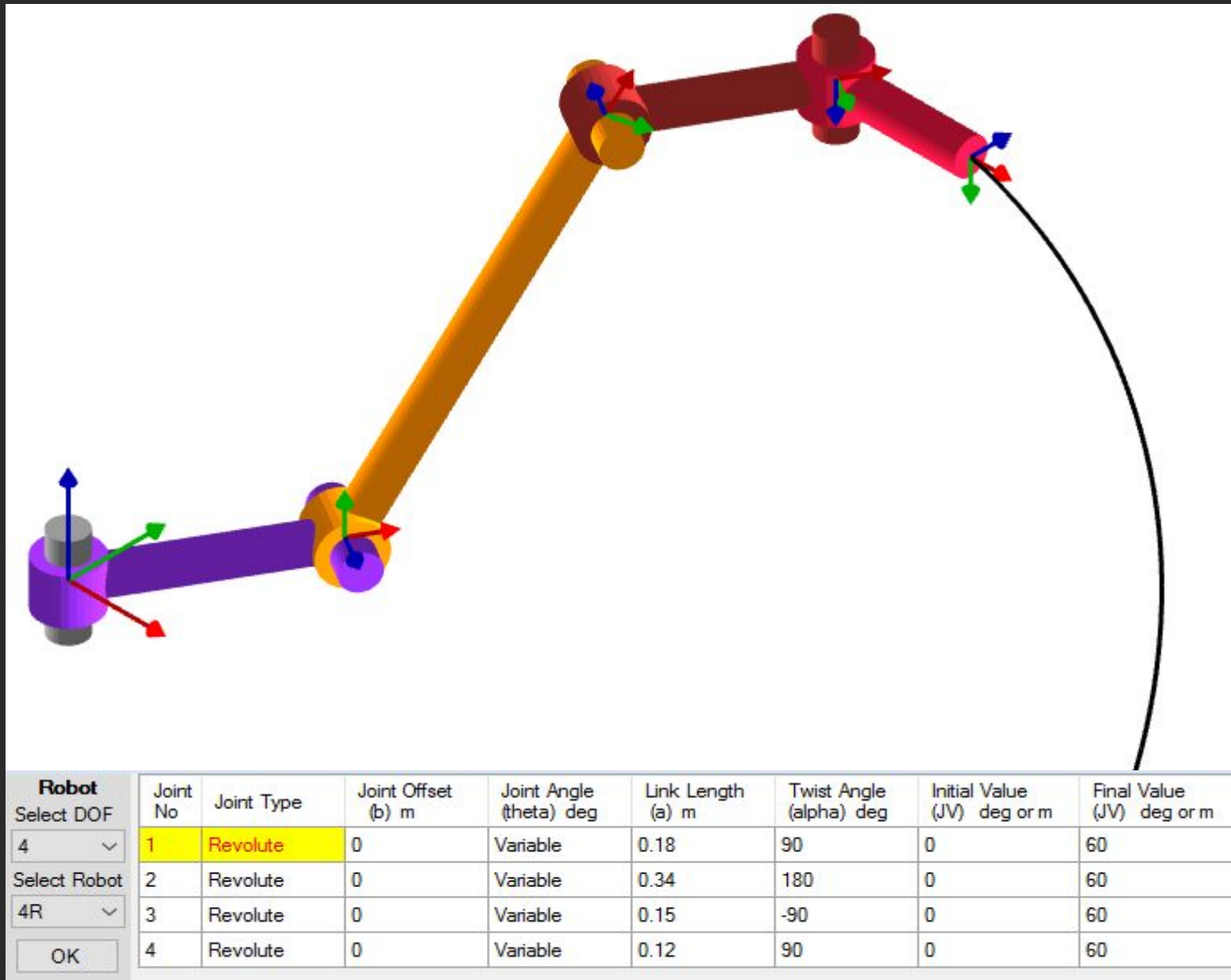
}

```

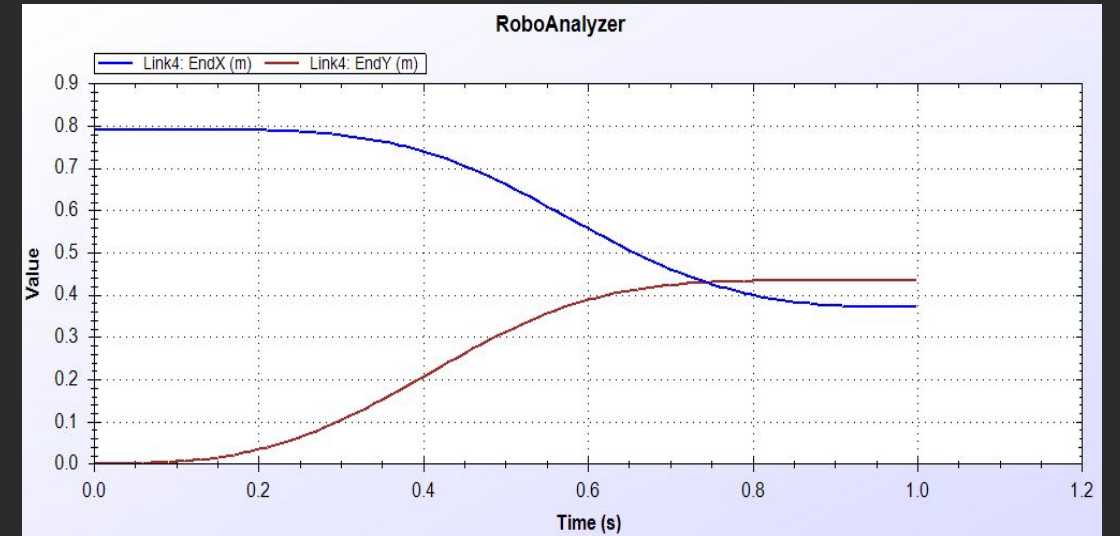
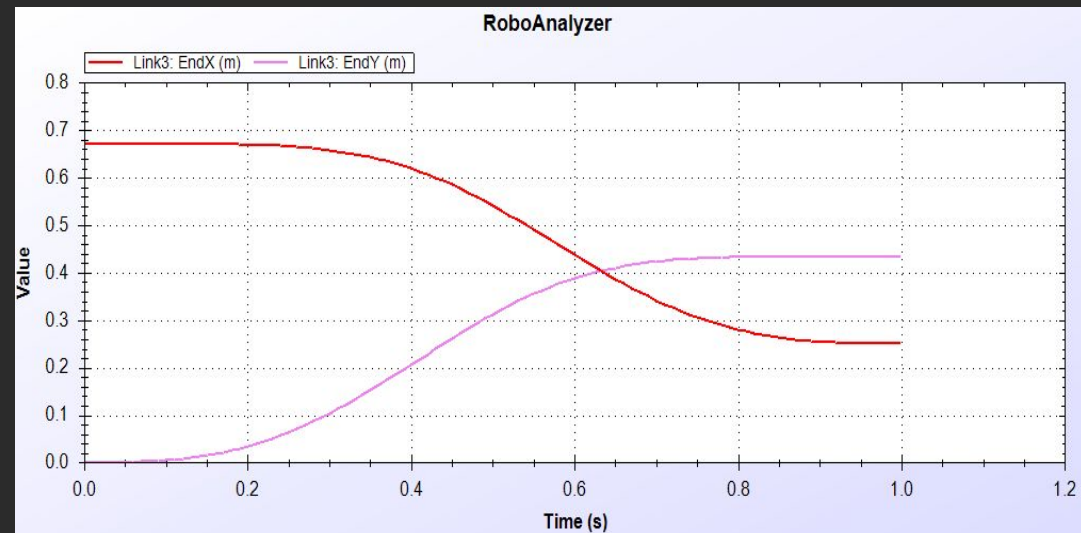
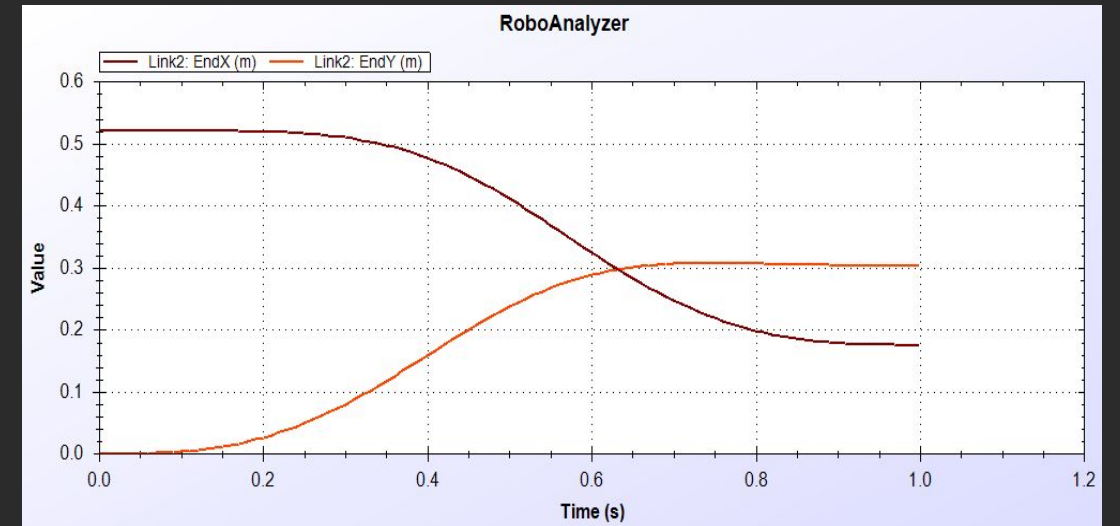
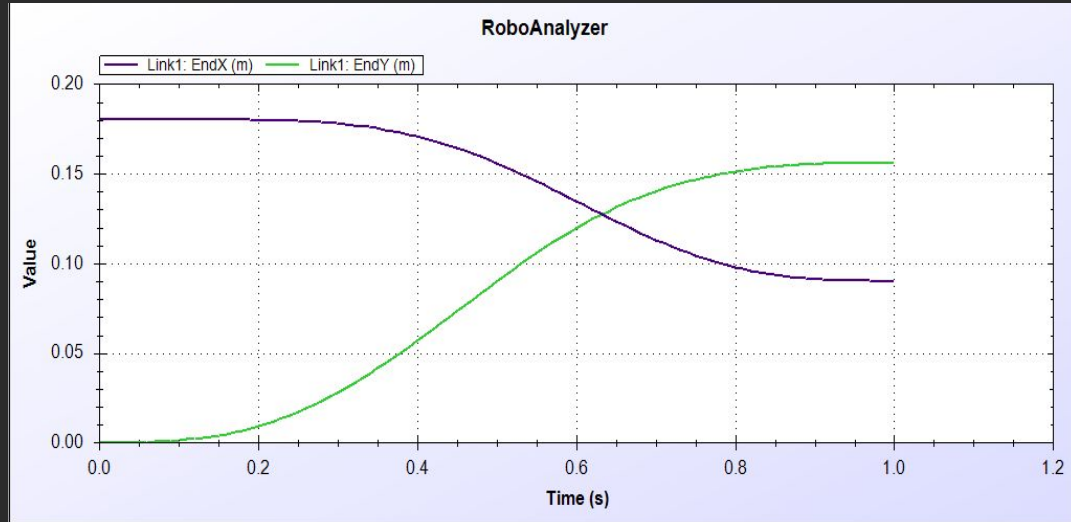
RESULTS

- ❖ Design of Elevator according to calculations
- ❖ Testing of voice recognition module and coupling with elevator mechanism
- ❖ Complete calculations of Manipulator (Workspace, Forward Kinematics, Dynamics)
- ❖ Hardware Completion
- ❖ Hand detection through color recognition module

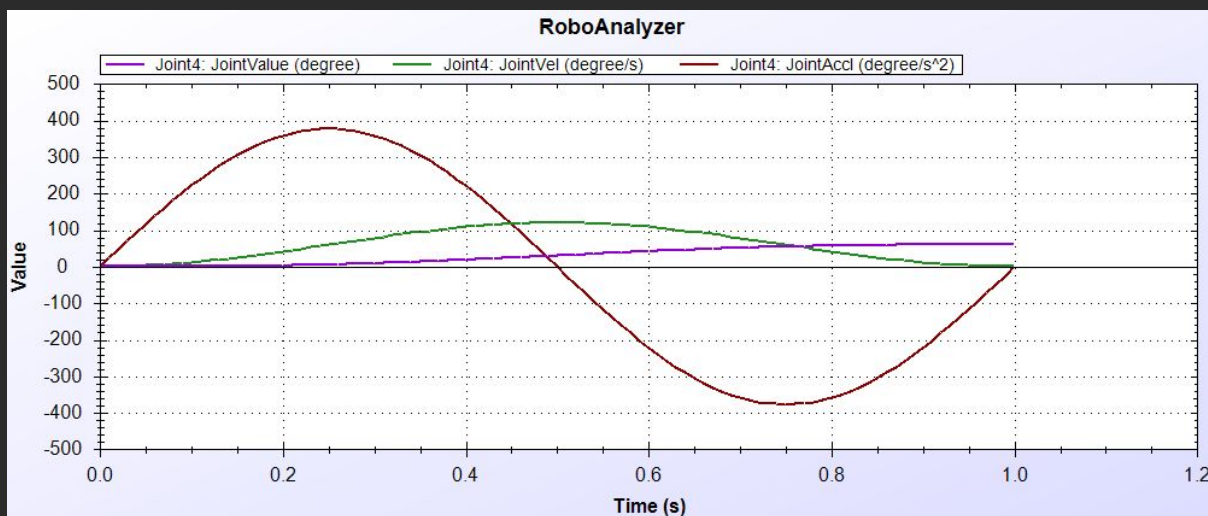
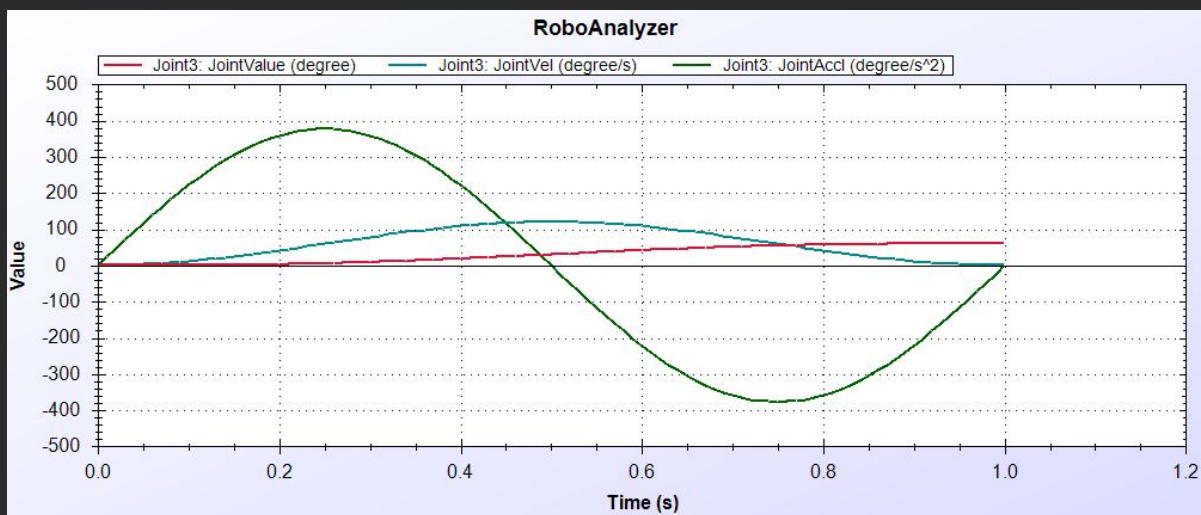
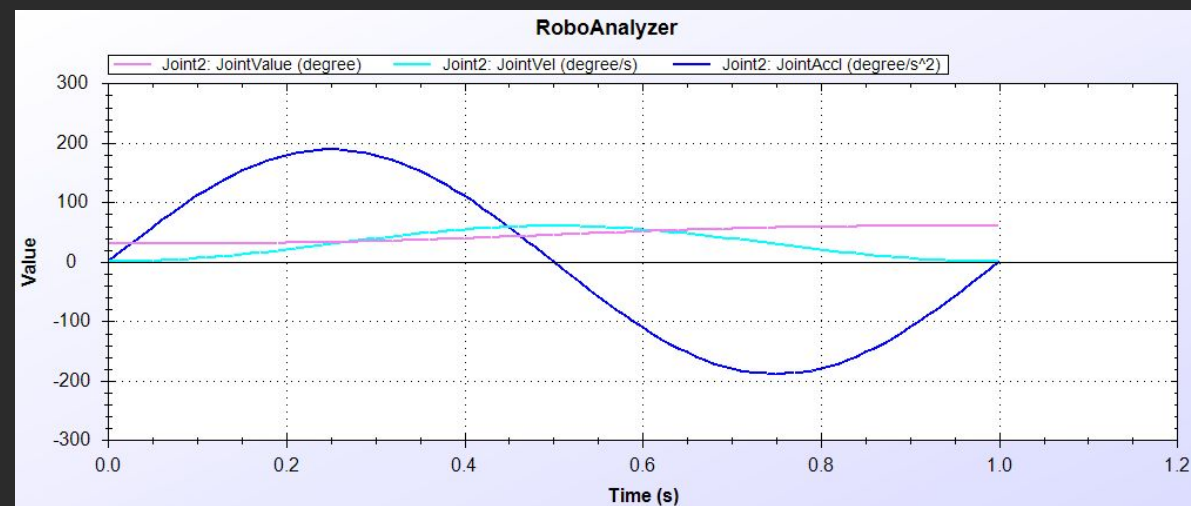
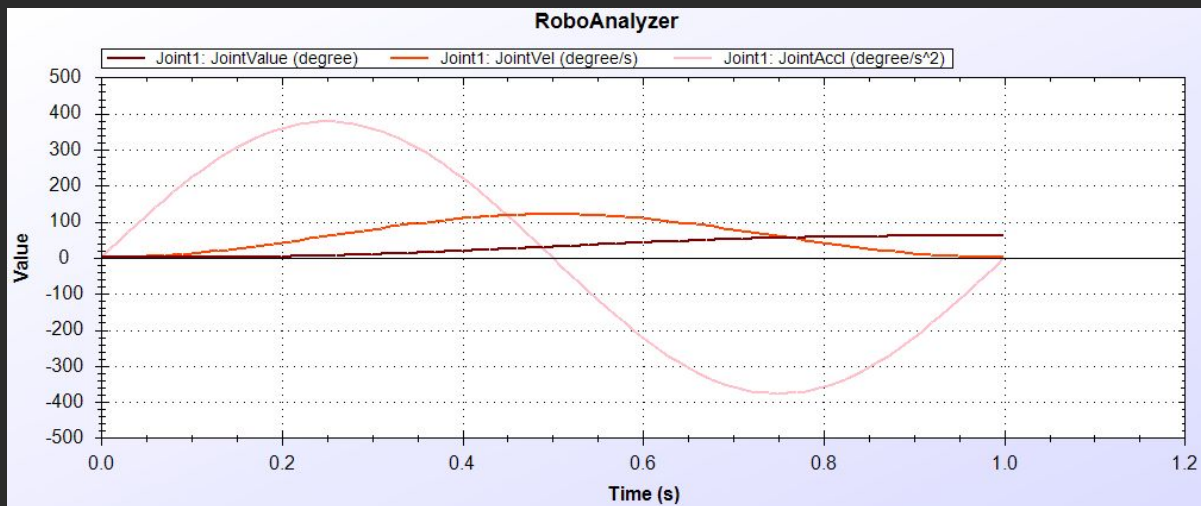
RESULTS FROM ROBOANALYZER



GRAPHS FROM ROBOANALYZER



GRAPH FROM ROBOANALYZER



Gantt Chart

■ Start Month ■ Completion period



REFERENCES



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- [2] Design and kinematic analysis of articulated manipulator | Elias Eliot, Deepak | 2015
- [3] Development of Dual arm assistant robot: The manipulator arm control system | Ivan, Arkady, Pavel | 2017
- [4] Survey of Robotic arm and Parameters | Virendra, Ritu | 2016
- [5] Color detection in RGB modeled images using MATLAB | Mary, Sudharshan | 2018
- [6] An Ergonomic assesment of operating table and surgical tool heights for seated otolaryngology procedures | Anam F. Azimuddin, Erik K. Wietzel, Kevin C McMains, Philip G. Chen | 2017

THANK YOU

FOR YOUR PRECIOUS TIME